



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM - 638 401

Module Test - II - May - 2026

II Semester

22GE003 & BASICS OF ELECTRICAL ENGINEERING

1	(i)	A proton moves in a region where both electric and magnetic fields are present. Engineers analyze how each field affects the motion of the particle. Determine why electric fields can change the speed of a charged particle while magnetic fields mainly change the direction of motion. (3 Marks - [U/P, 1])
2	(ii)	A charged particle moves perpendicular to a magnetic field. Formula Hint $F = q \times v \times B$ Determine how doubling the magnetic field affects the magnetic force. (2 Marks - [U/C, 1])
3	(i)	Scientists use crossed electric and magnetic fields in instruments such as mass spectrometers to control particle motion. Determine why velocity selectors are important in such instruments. (3 Marks - [U/P, 1])
4	(ii)	A velocity selector has electric field 2000 N/C and magnetic field 0.02 Tesla. Formula Hint $v = E / B$ Determine the velocity of particles that pass through undeflected. (2 Marks - [U/C, 1])
5	(i)	A charged particle enters a magnetic field with its velocity parallel to the direction of the magnetic field. Engineers studying electromagnetic forces analyze the resulting magnetic force acting on the particle. Formula Hint $F = q \times v \times B \times \sin\theta$ Determine the magnetic force acting on the particle and explain the result. (3 Marks - [U/P, 1])
6	(ii)	A charge 4×10^{-6} C is placed in an electric field of 2500 N/C. Formula Hint $F = q \times E$ Determine the electric force acting on the charge. (2 Marks - [U/C, 1])
7	(i)	In a velocity selector experiment, charged particles enter a region where electric and magnetic fields are perpendicular to each other. Some particles pass straight through the device without deflection while others deviate from the path. Determine why only particles with a specific velocity pass through the selector without deflection. (3 Marks - [U/P, 1])

8	(ii)	A velocity selector uses crossed electric and magnetic fields. Formula Hint $v \propto E$ $v \propto 1/B$ Determine how doubling the magnetic field affects the selected velocity. (2 Marks - [U/C, 1])
9	(i)	A particle with charge 4×10^{-6} C is placed in a uniform electric field of 5×10^3 N/C between two conducting plates. Engineers analyzing electrostatic forces must determine the magnitude of the force acting on the particle and understand how the electric field controls particle motion. Formula Hint $F = q \times E$ Determine the electric force acting on the charge and analyze how increasing the charge would affect the force. (5 Marks - [Ap/P, 2])
10	(i)	A conductor of length 0.30 m carrying a current of 5 A is placed in a uniform magnetic field of 0.40 Tesla. The conductor is positioned perpendicular to the magnetic field. Engineers studying electromagnetic devices must determine the force acting on the conductor and understand how magnetic fields produce mechanical motion in electric motors. Formula Hint $F = B \times I \times L$ Determine the force acting on the conductor and analyze how increasing the magnetic field strength affects the force. (5 Marks - [Ap/P, 2])
11	(i)	An electron moves inside a region where a uniform electric field of 3×10^4 N/C exists between two plates. Engineers analyzing particle accelerators must determine the electric force acting on the electron. Charge of electron = 1.6×10^{-19} C Formula Hint $F = q \times E$ Determine the magnitude of the electric force acting on the electron. (5 Marks - [Ap/P, 2])
12	(i)	A straight conductor carrying current is placed in a magnetic field and experiences a force due to the interaction between moving charges and the magnetic field. This phenomenon is widely used in electric motors where electrical energy is converted into mechanical motion. Formula Hint $F = B \times I \times L$ Explain how the interaction between current and magnetic field produces a force on the conductor and how this principle enables the operation of electric motors. (5 Marks - [Ap/C, 2])
13	(i)	A proton with charge 1.6×10^{-19} C enters a uniform magnetic field of 0.03 Tesla with velocity 3×10^6 m/s perpendicular to the field direction. The magnetic field exerts a force that causes the proton to move in a circular path. Engineers studying cyclotron devices must determine the magnetic force acting on the proton. Formula Hint Magnetic Force $F = q \times v \times B$ Determine the magnetic force acting on the proton and interpret how increasing the magnetic field strength affects the force. (5 Marks - [Ap/P, 2])

14	(i)	Two long parallel conductors are placed 0.05 m apart in a laboratory setup used to study electromagnetic interaction between current carrying wires. The first conductor carries a current of 8 A, while the second conductor carries 6 A. A segment of length 0.50 m of the conductors is considered for analysis. Engineers must determine the magnitude of the force between the conductors in order to understand the magnetic interaction between current carrying wires used in power transmission systems. Formula Hint Force between parallel conductors $F = (\mu_0 \times I_1 \times I_2 \times L) / (2\pi \times d)$ $\mu_0 = 4\pi \times 10^{-7}$ Determine the force between the conductors and interpret whether the conductors attract or repel if both currents flow in the same direction. (5 Marks - [Ap/P, 2])
15	(i)	A charged particle enters a magnetic field at an angle to the field direction. Because the velocity has components both perpendicular and parallel to the magnetic field, the particle follows a helical trajectory in the magnetic field. Engineers analyzing magnetic confinement systems must understand the relationship between particle motion and magnetic force. Formula Hint $F = q \times v \times B \times \sin\theta$ Explain how the magnetic force influences the motion of the particle and why the particle follows a helical path. (5 Marks - [Ap/C, 2])
16	(i)	In an experiment involving two long parallel conductors carrying electric currents, each conductor produces a magnetic field around it. This magnetic field interacts with the current in the other conductor and produces a force between them. Engineers must analyze the physical mechanism responsible for this interaction. Formula Hint Magnetic interaction principle $F \propto I_1 \times I_2$ $F \propto 1 / d$ Explain how the magnetic field produced by one conductor exerts a force on the other conductor and why the direction of current determines whether the wires attract or repel. (5 Marks - [Ap/C, 2])
17	(i)	In a laboratory demonstration, a transmitter produces electromagnetic waves that propagate through space and reach a detector placed at some distance. The detector records the arrival of energy even though there is no direct electrical connection between the transmitter and the detector. Establish how energy is transferred between the transmitter and detector. (3 Marks - [U/P, 2])
18	(ii)	Solar energy travels from the Sun to Earth through empty space. Identify the mechanism responsible for this energy transfer. (2 Marks - [U/C, 1])
19	(i)	A metal plate moving between the poles of a magnet experiences resistance to its motion. Relate this effect to the formation of eddy currents. (3 Marks - [U/P, 2])
20	(ii)	A metallic conductor placed in a changing magnetic field develops circulating electric currents within it. Identify these currents. (2 Marks - [U/C, 1])

21	(i)	During a microwave transmission experiment, energy generated by a transmitter travels through space and is detected by a receiver placed several meters away. Relate this observation to energy transfer in electromagnetic fields. (3 Marks - [U/P, 2])
22	(ii)	During an experiment on wireless communication, a transmitter generates electromagnetic waves that propagate through space and reach a receiver placed several meters away. Even though there is no physical conductor connecting the transmitter and receiver, the receiver successfully detects the signal energy carried by the wave. Determine the phenomenon responsible for this energy transfer. (2 Marks - [U/C, 1])
23	(i)	During a laboratory demonstration, a copper disc rotates between the poles of a magnet. Students observe that the disc gradually slows down even though no mechanical friction is applied. The instructor explains that the magnetic field induces circulating currents inside the metal disc which oppose its motion. Establish how eddy currents influence the motion of conductors in magnetic fields. (3 Marks - [U/P, 2])
24	(ii)	In a classroom experiment, a teacher moves a strong magnet close to a thick aluminum plate. Students notice that the plate slightly resists the motion of the magnet. The teacher explains that this happens because changing magnetic fields induce circular electric currents inside the conductor. Determine the phenomenon responsible for this effect. (2 Marks - [U/C, 1])
25	(i)	In power electronic systems, inductors are commonly used to store energy temporarily in magnetic fields. Engineers design inductors so that when current passes through them, energy is stored in the magnetic field created around the coil. This stored energy can later be released when the current changes, helping regulate electrical circuits. Evaluate the role of magnetic fields in storing electrical energy in inductors. (5 Marks - [Ap/P, 2])
26	(i)	In an electrical laboratory, a coil connected to a power supply produces a magnetic field around it when current flows. When the current in the coil suddenly changes, the magnetic field around it also changes. This change in the magnetic field induces an electromotive force within the same coil that opposes the change in current. Engineers recognize this behavior as an important property of coils used in electrical circuits. Analyze the concept of self inductance in an electrical coil. (5 Marks - [An/P, 2])
27	(i)	In an electrical engineering laboratory, students connect a coil to a power supply and observe the behavior of the magnetic field produced around the coil. As current flows through the coil, a magnetic field develops around it and energy becomes stored within that magnetic field. When the power supply is switched off, the magnetic field collapses and releases the stored energy back into the circuit. Analyze how energy is stored in a magnetic field produced by an electric current. (5 Marks - [Ap/P, 2])

28	(i)	During a circuit experiment, a student observes that when the current through a coil is suddenly increased, the coil briefly resists the change in current. This happens because the magnetic field associated with the coil changes and produces an induced electromotive force that opposes the change. Assess how self inductance affects current changes in electrical circuits. (5 Marks - [Ap/P, 2])
29	(i)	In electrical power generators, mechanical rotation of coils within magnetic fields produces electric current that is supplied to homes and industries. As the coil rotates, the magnetic field linking the coil continuously changes. This changing magnetic field produces an induced electromotive force that drives current through the external circuit. Evaluate how electromagnetic induction enables electricity generation in generators. (5 Marks - [Ap/P, 2])
30	(i)	In space research experiments, scientists use powerful laser beams to manipulate tiny particles. When the laser beam strikes a microscopic particle, it pushes the particle slightly in the direction of the beam. This happens because the electromagnetic field of the laser carries momentum that is transferred to the particle upon interaction. Evaluate how electromagnetic radiation transfers momentum to matter. (5 Marks - [Ap/P, 2])
31	(i)	In transformer systems used in power distribution, alternating current flowing through one coil produces a changing magnetic field. This varying magnetic field interacts with a nearby coil and induces an electromotive force in it. As a result, electric energy is transferred between the two coils without direct electrical contact. Assess how changing magnetic fields produce induced emf in nearby conductors. (5 Marks - [Ap/P, 2])
32	(i)	Scientists studying solar radiation observe that sunlight not only carries energy but also exerts a small pressure on surfaces it strikes. Sensitive instruments detect that when electromagnetic waves from the Sun hit a reflective surface, a tiny force is produced. This force indicates that electromagnetic radiation carries momentum even though it consists only of electric and magnetic fields traveling through space. Analyze how electromagnetic fields can carry linear momentum. (5 Marks - [An/P, 2])